

Nonbacterial Thrombotic Endocarditis

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 Adapted from: [Nonbacterial Thrombotic Endocarditis](#)

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Learning Objectives (4)

Versions

After completing this brick, you will be able to:

- 1 Differentiate the clinical presentation of nonbacterial thrombotic endocarditis (NBTE).
- 2 Explain the pathophysiology of NBTE.
- 3 Interpret clinical findings to properly diagnose NBTE.
- 4 Outline the management plan of NBTE.



CASE CONNECTION

MB, a 46-year-old patient with systemic lupus erythematosus, comes to your office because she has seen blood in her urine for 2 days. “First, I noticed a deep pain in my back, on the left side, right above my bottom. Then, I noticed the blood.” MB’s vital signs are within normal limits. Her physical exam is unremarkable except for a new holosystolic murmur at the apex of her heart that radiates to her axilla.

What could be causing MB’s new murmur? How is the murmur associated with her hematuria? Consider your answers as you read, and we’ll revisit MB at the end of the brick.

[GO TO CONCLUSION](#) ↓

What Is Nonbacterial Thrombotic Endocarditis?

Endocarditis is inflammation of the inner lining of the heart (the endocardium) and/or the heart valves. Although endocarditis is most often caused by an infection (AKA infective endocarditis), it can uncommonly develop in the absence of an infection, producing what are known as sterile vegetations. This disease process is known as **nonbacterial thrombotic endocarditis (NBTE)**. NBTE is rare and usually occurs in patients with an underlying hypercoagulable state, such as malignancy, systemic lupus erythematosus (SLE), or antiphospholipid syndrome (APS).

A hypercoagulable state underlies most NBTE.

How Do Patients with Nonbacterial Thrombotic Endocarditis Present? (LO #1)

Patients with NBTE are often asymptomatic. In fact, it is most often diagnosed at autopsy, detected as an incidental finding. There may be no indications of valvular or heart dysfunction. This is because since the valvular lesions are non-infectious, they rarely result in destruction of the valve, as commonly occurs in infective endocarditis.

When present, symptoms are typically because of thromboembolism rather than due to direct valvular damage. In some cases, patients may not be diagnosed until after emboli have occurred. Common sites of embolization include the spleen, kidney, skin, and extremities that may present as flank pain, hematuria, rash, and digital ischemia. However, emboli to the central nervous system and coronary arteries are more likely to reveal the diagnosis due to the presence of more worrisome symptoms (i.e. stroke, delirium, chest pain).

stroke.

Patients with severe NBTE often present with embolic symptoms like

What Is the Pathophysiology of Nonbacterial Thrombotic Endocarditis? (LO #2)

Let's look at why vegetations form in NBTE. Most patients have a pre-existing **hypercoagulable state**.

Malignancy

Malignancy is associated with thrombosis, due to an activation of the coagulation system as cancer cells produce procoagulant substances, as well as cytokines that can damage the endothelium.

Systemic Lupus Erythematosus

Systemic lupus erythematosus (SLE) is a multisystem autoimmune disease that can show symptoms in many organ systems, including the heart (pericarditis, or less often NBTE). The mechanism of vegetation formation is not well defined. Immune complex formation has been implicated, as the vegetations have been shown to contain immune complexes and cellular debris.



CLINICAL CORRELATION

Some pathologists separate the sterile endocarditis of SLE into a separate diagnosis termed **Libman-Sacks endocarditis**, but this is not uniformly agreed upon. Others use the eponym (Libman-Sacks endocarditis) for a form of NBTE with very large thrombi.

Antiphospholipid Syndrome

Antiphospholipid syndrome (APS) occurs when autoantibodies attack cell membranes, causing excess clot formation. APS can occur as an idiopathic disorder but also in association with SLE or infections.

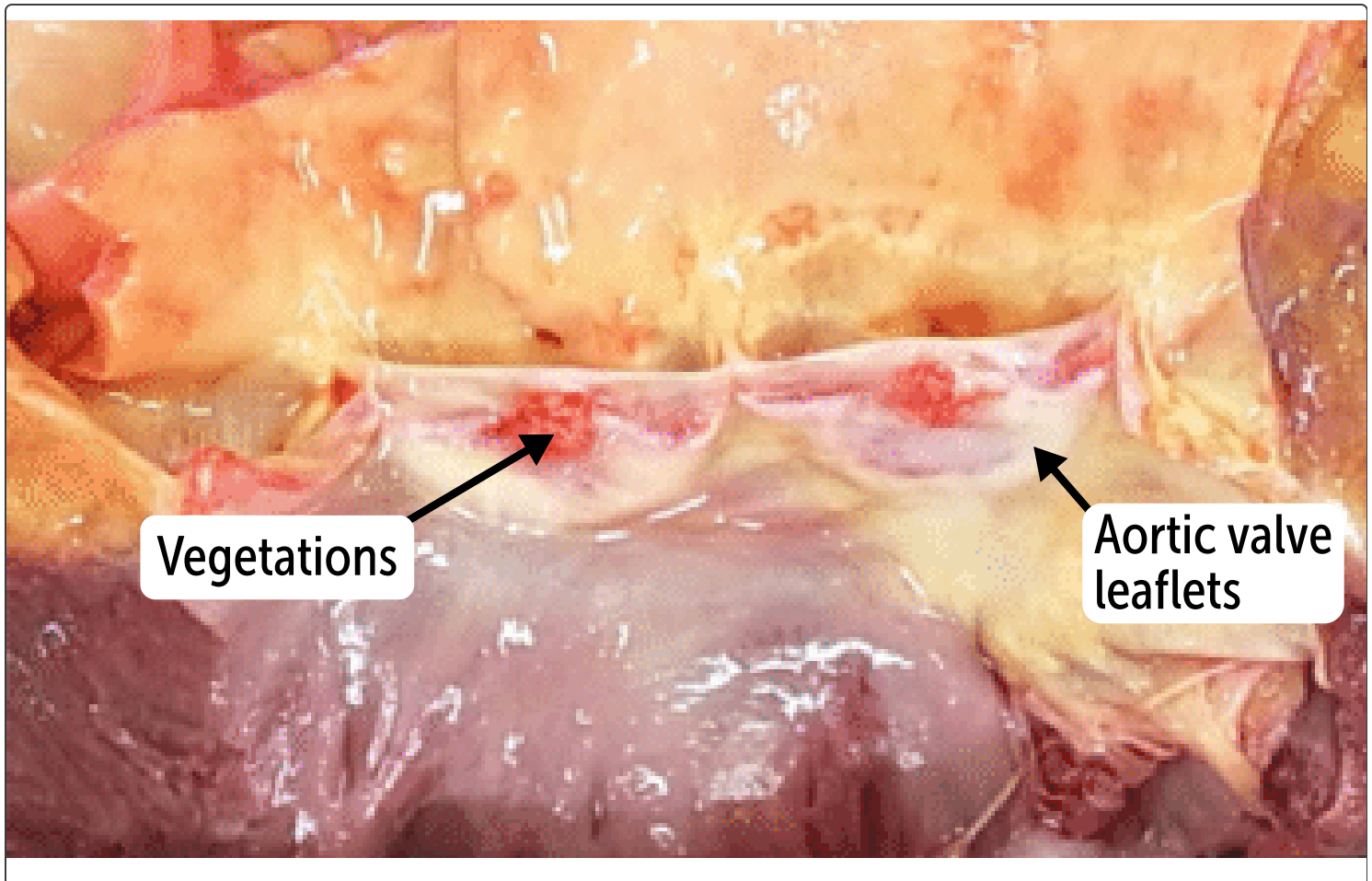
can damage the endothelium.

hypercoagulable state associated with production of cytokines that

Patients with malignancies are prone to NBTE because of their

Gross Pathology

Sterile vegetations are just that—sterile—meaning there are no infectious organisms (i.e., bacteria or fungi) in the thrombi. The sterile vegetations of NBTE range in size from large, warty growths called verrucous vegetations (*verruca* means *wart*) down to small, rounded growths (Figure 1).



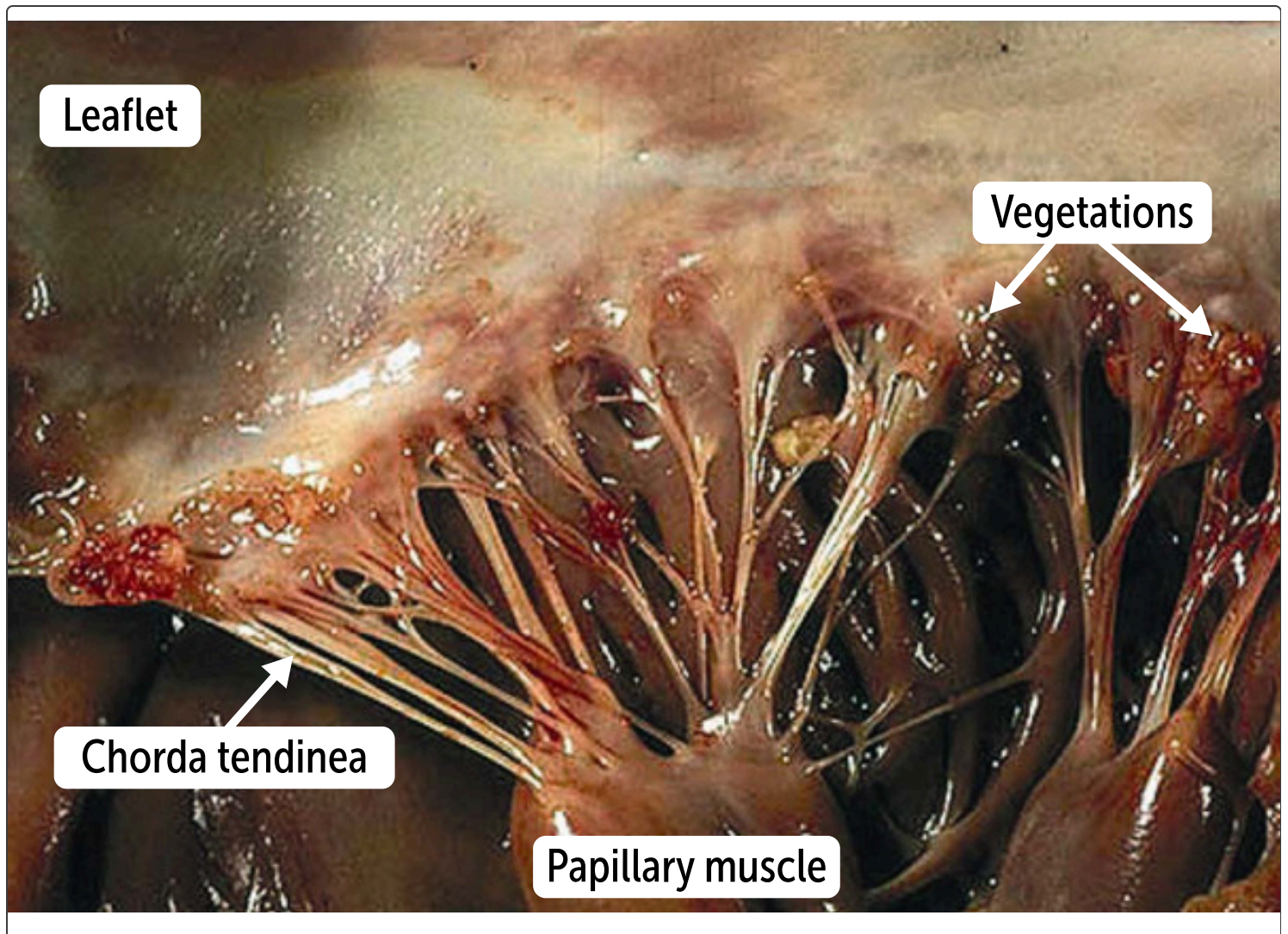
QUIZ

Tap image for quiz

Figure 1. Valve Vegetations in NBTE.

The vegetations can occur on any valve, but vegetations associated with SLE favor the mitral and aortic valves. Single or multiple growths can be seen

near the margins of the valve cusps. Some large thrombi can form on both sides of the valve, as in SLE endocarditis (Figure 2).



QUIZ

 Tap image for quiz

Figure 2. Vegetations on the Valve Leaflet and Chordae Tendineae.

A comparison of the appearance of vegetations between IE, NBTE, and Libman-Sacks endocarditis is provided in Figure 3.

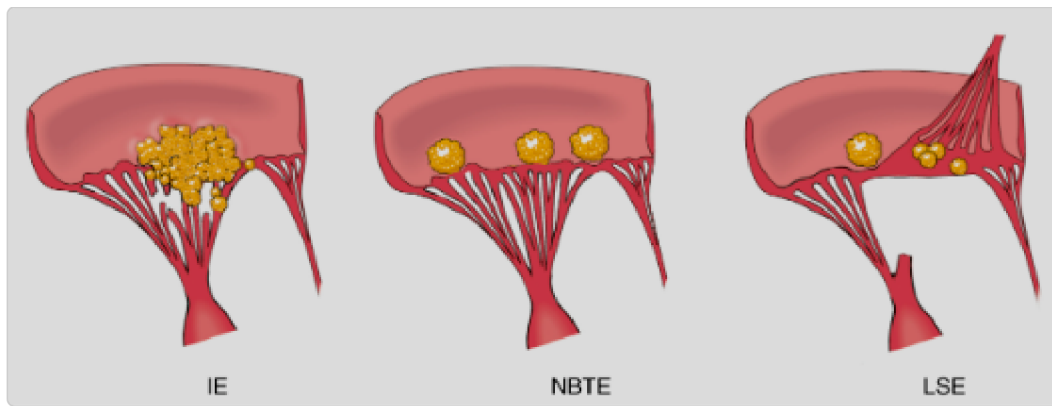


Figure 3. Infective endocarditis (IE) is characterized by large, irregular, often destructive vegetations. Nonbacterial thrombotic endocarditis (NBTE) typically manifest with small- to medium-sized bland nondestructive vegetations. Libman-Sacks endocarditis (LSE) is characterized by small- to medium-sized vegetations that can be attached to either side of the valve leaflets.

Complications

Vegetations in NBTE have the potential to **embolize**. Because they may be loosely adherent to the valves, they are likely to fall off the valves and embolize to other parts of the body. One of the most common places for embolization is the brain (manifesting as stroke), but the heart, kidney, and spleen are also commonly affected.

The sterile vegetations of NBTE are rarely destructive to the valve, unlike those of infective endocarditis (IE). However, sometimes inflammatory valvulitis can impair valve function, potentially causing valve regurgitation.

The lesions of Libman-Sacks endocarditis characteristically heal with scarring; the ensuing fibrosis can result in lesions that resemble chronic rheumatic heart disease.

complication of NBTE.

Embolization to the brain or other body organs is the most dangerous

How Do We Diagnose Nonbacterial Thrombotic Endocarditis? (LO #3)

The diagnosis of NBTE is difficult. Patients are often asymptomatic, and physical examination is often normal. So, patients may be diagnosed incidentally or after their disease has progressed and/or complications have occurred.

Echocardiography

Anyone presenting with a new heart murmur or signs and symptoms of new-onset heart failure should get an **echocardiogram**, the best test to detect valvular vegetations. Echocardiography cannot distinguish between infectious and sterile vegetations, so blood cultures should be drawn if a vegetation is seen, since not all patients with infective endocarditis have fever.



CLINICAL CORRELATION

Some cases of infective endocarditis are caused by bacterial organisms that are difficult to isolate with blood cultures - so called culture negative endocarditis. In a patient with valvular lesions identified on echocardiogram but negative blood cultures, it can be challenging to separate NBTE from culture negative IE. In these instances, the clinical context, patient history, and additional laboratory investigation (i.e., inflammatory or serologic markers) are paramount to arriving at the correct diagnosis.

A patient with fever, vegetations, and negative blood cultures with known mitral valve prolapse who recently underwent a dental procedure is more likely to have culture negative IE than a non-febrile patient who has a known history of antiphospholipid antibody syndrome but no known pre-existing cardiac abnormalities (who is more likely to have NBTE).

vegetations.

Echocardiograms are the best test to detect valvular disease and

Laboratory Tests

Three sets of blood cultures should be done to exclude infective endocarditis. If autoimmune disease (SLE and APS) is suspected as the cause, appropriate laboratory workup such as an antinuclear antibody (ANA), anti-Smith, rheumatoid factor, anti-dsDNA, and antiphospholipid antibody panel should also be done.

How Do We Treat Nonbacterial Thrombotic Endocarditis? (LO #4)

Treatment involves identifying the underlying cause (malignancy, SLE, APS) and treating that if possible. Unless a contraindication exists, these patients should be anticoagulated with heparin (unfractionated or low molecular weight), then coumadin to prevent emboli.



CASE CONNECTION

[BACK TO INTRODUCTION](#) ↑

Thinking back to MB, what is causing her new murmur, and how does the murmur relate to her hematuria?

The presence of a new murmur in a patient with SLE is a very significant finding because it raises suspicion for noninfective endocarditis. You suspect that a noninfective vegetation has formed on MB's mitral valve, causing her new murmur, and that a portion has embolized to her kidney, causing her flank pain and hematuria. You order an echocardiogram to confirm your diagnosis.

Summary

- Nonbacterial thrombotic endocarditis (NBTE) refers to inflammation and subsequent thrombosis on the surface of heart valves in the absence of an infection.
- It's rare and usually occurs in patients with an underlying hypercoagulable state.
- Patients are usually asymptomatic. If present, symptoms include embolic symptoms like neurologic findings from stroke.
- NBTE frequently occurs in patients with an underlying hypercoagulable state.
- Common causes are cancers, systemic lupus erythematosus (SLE), and antiphospholipid antibody syndrome (APS).
- Vegetations in NBTE are most often small, sterile, and nondestructive in contrast with the destructive vegetations of infective endocarditis.
- Vegetations in NBTE may embolize to different parts of the body.
- Diagnosis is made by echocardiography.
- Blood cultures negative for infective endocarditis support the diagnosis.
- Treatment is of the underlying cause (eg, SLE, APS, malignancy).
- Anticoagulation should be used if no contraindication exists.

Review Questions

1. Which of the following mechanisms most commonly leads to nonbacterial thrombotic endocarditis?

- A. Autoimmune response against the pericardium
- B. Colonization of the heart valve by *Staphylococcus aureus*
- C. Hypercoagulable state
- D. Malignant cancer metastasizing to the heart valves
- E. Structural valvular disease

Explanation (requires correct answer)

2. Which of the following is a common predisposing factor associated with nonbacterial thrombotic endocarditis?

- A. IV drug use
- B. Malignancy

- C. Polyarteritis nodosa
- D. Progressive systemic sclerosis
- E. Rheumatic heart disease

Explanation (requires correct answer)

3. A patient previously diagnosed with systemic lupus erythematosus comes to the clinic with history of sudden-onset weakness of her right leg. A murmur is auscultated on cardiac examination that was not documented in previous visits. What is the most likely complication that this patient may have developed?

- A. Acute infective endocarditis
- B. Brain embolization
- C. Cardiomyopathy

- D. Septic shock
- E. Valvular degeneration

Explanation (requires correct answer)

4. What is the best clinical test to diagnose valvular vegetations?

- A. Anti-neutrophil cytoplasmic antibodies (ANCA's)
- B. Blood cultures
- C. Chest x-ray
- D. D-dimers
- E. Echocardiogram

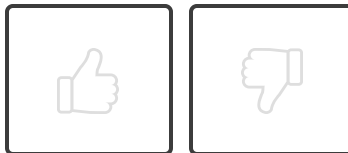
Explanation (requires correct answer)

5. The management of NBTE involves treatment meant to prevent which known complication?

- A. Diastolic heart failure
- B. Valve destruction
- C. Embolization
- D. Myocarditis
- E. Pleural effusion

Explanation (requires correct answer)

How would you rate this Brick?



References

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01** Differentiate the clinical presentation of nonbacterial thrombotic endocarditis (NBTE).
- 02** Explain the pathophysiology of NBTE.
- 03** Interpret clinical findings to properly diagnose NBTE.
- 04** Outline the management plan of NBTE.

Objective Confidence: 0/4